

BB 10 to integrate Evernote

Reports suggest that Evernote – the online notes service – will be deeply integrated in the upcoming BlackBerry 10 OS



Modders rejoice!

Budding PC modders, check out the list of Top 10 tools used for case modding <http://dgit.in/UTDg2J>

Motherboard Test

HOW WE TESTED

We called for mini-ITX boards, as we were looking at boards that will not only make for a good home-theatre PC, but can actually be used as a proper system for performing your everyday tasks as well. The common components used were as follows:

Processor: Intel Core i7-3770K (for most Intel Z77/H77/H61 boards); AMD A8-3850 (for A75 boards) and Intel Core i5-2500K (for Intel DH61DL)

RAM: 2x4GB Kingston HyperX DDR3 for DIMM slots; 4GB Kingston RAM for SO-DIMM slots

SSD: Kingston HyperX 120GB

Graphics Card: None

Monitor: Dell S2750L

Power Supply Unit: Corsair HX 620W

Operating System: Windows 7 Ultimate (64-bit)

We put a lot of stress on the features offered by the boards as we feel the board's prowess is a function of its components. We were looking at boards that can be used for making your mainstream systems. Boards with USB 3.0 and on-board Wi-Fi got more points. Similarly boards which had better heat-sinks around VRM modules and better options to overclock got a slight edge over the others. Bundled accessories were also taken into consideration.

The layout of the board was scored, we checked if there was enough spacing around the CPU region to comfortably attach the CPU cooler and which way the SATA ports were pointing (those pointing up got a lesser score). Spacing between the RAM slots and the graphics card was checked. For build quality we noted the sturdiness of the heat sinks around

the VRM regions and the PCH. Motherboards lacking a heatsink around the VRM region got lesser score. Board with all solid state capacitors got a higher score. Back panel I/O ports were noted and also the option to expand USB ports. Fan connectors and SATA connections were noted.

We ran the boards through our standard set of benchmark tests. As with most of our motherboard tests, these tests are only run to give you an indication and we do not give any weightage to these scores in our final evaluation of the boards, as we notice that for similar chipsets, the variation in scores is miniscule.

We did a thorough checking and testing (wherever applicable) of the feature set offered by a motherboard and based on the price have decided on the winners.

chipset and the PCIe x16 slot. This is thanks to the extra PCB – the daughterboard – which houses the Digi+ VRM (voltage regulator module) circuitry. The idea to add in another PCB on top of the ITX board was to get rid of the space-constraints of the mini-ITX form factor, and at the same time give the user a power-packed board. The daughterboard is 4.2 cm in height, rising at a right angle to the P8Z77-I board. Not only does it house the solid state capacitors but also a passive heatsink. It is an intelligent layout as by taking the VRM circuitry on the daughter-board, there is enough space around the CPU socket for adding in a fan. On the lower side of the boards, things are pretty dense, with a PCIe 3.0 slot running at x16 speeds, packed very close to the chipset which has four SATA ports (2xSATA 2 and 2xSATA 3) on the right and the Wi-Fi and bluetooth card on the left-hand side.

When it comes to heat-sinks, all the three Z77 boards are impressive. The heat-sink on the ZOTAC board is quite good, with the VRM heatsink

connected to the chipset heat-sink via a heat pipe helping in quicker heat dissipation. The ASRock heat-sink is not as elaborate as the ones seen on ZOTAC & ASUS but is decent enough. All the three boards have the SATA ports pointing upwards, with a 2x2 arrangement on ZOTAC and ASRock boards and a 1x4 arrangement on the ASUS board. There are two USB 2.0 headers and a USB 3.0 header on all boards. All the boards come with on-board Wi-Fi card which fits in the mini PCIe slot. Out of the box, the wire management of the Wi-Fi card was the best on ASUS and ASRock boards where it was nicely manoeuvred through the components on the board. The ZOTAC board's wire was found to be dangling. These are little



Gigabyte H77N-WIFI

things, but it shows the attention to detail by the board makers.

The ZOTAC board was the only one which had dedicated power and reset buttons on board along with a POST error-code readout LED. These are desirable features in a board. The ASUS P8Z77-I offers the MemOK button which helps to check on faulty memory modules, and on the back I/O panel it features the clear CMOS and BIOS flash buttons. The region around the

CPU was the best in the ASUS board followed by ASRock and ZOTAC boards. The 8-pin CPU power connector found on all three boards is placed most strategically on the ASUS board. The back I/O panels on all the boards are quite well represented. Both ASUS and ASRock offer four USB 3.0 ports and sport a DVI, HDMI and a DisplayPort each whereas the ZOTAC board sports two HDMI ports along with one mini-DisplayPort. It